

1.

Exam Section
■ Mark

National Board of Medical Examiners
Surgery Self-Assessment

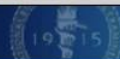
Four days after undergoing coronary artery bypass grafting, a hospitalized 47-year-old man has the sudden onset of severe pain in his left great toe. He was admitted to the hospital 10 days ago for evaluation of chest pain with exertion, and received aspirin and heparin therapy for 3 days. Cardiac catheterization on admission showed coronary artery disease involving three vessels. The operation and initial postoperative course were uncomplicated. Vital signs are within normal limits. Examination shows a tender, cyanotic left great toe. There are new ecchymoses over the trunk and upper and lower extremities. The surgical incision appears clean and intact. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hematocrit	37%
Leukocyte count	12,200/mm ³
Platelet count	8000/mm ³
Prothrombin time	12 sec (INR=1)
Partial thromboplastin time	18 sec

Which of the following is the most likely diagnosis?

- ☐ A) Cholesterol embolization syndrome
- ☐ B) Disseminated intravascular coagulopathy
- ☐ C) Gout
- ☐ D) Heparin-induced thrombocytopenia
- ☐ E) Immune thrombocytopenic purpura

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A 42-year-old man is brought to the emergency department following a motor vehicle collision. He was driving at 25 mph when his car hit a tree. He was wearing a seat belt. He briefly lost consciousness at the scene. On arrival, he is dazed but conscious. He has a head laceration and bruising over the anterior abdominal wall in the region of the seat restraint. Laboratory studies and x-rays show no abnormalities. Three days later, he vomits bilious material that tests positive for blood. He is afebrile. His abdomen is nontender but distended and tympanic to percussion. A complete blood count, serum electrolyte concentrations, amylase activity, and liver function tests are within normal limits. A flat plate x-ray shows a large gastric bubble. A CT scan of the abdomen is most likely to show which of the following?

- ☐ A) Acute cholecystitis
- ☐ B) Contained splenic rupture
- ☐ C) Duodenal hematoma
- ☐ D) Intrahepatic hematoma
- ☐ E) Splenic hematoma

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Surgery Self-Assessment

A 42-year-old woman comes to the physician because of a 2-day history of right upper abdominal pain and generalized itching. During the past 2 months, she has had three episodes of similar symptoms associated with nausea. She has not had fever or vomiting. She has no history of serious illness and takes no medications. Her current temperature is 37.3°C (99.1°F), pulse is 94/min, and blood pressure is 124/78 mm Hg. Examination shows scleral icterus. A mildly tender mass is palpated in the right upper quadrant of the abdomen. There are no peritoneal signs. Her leukocyte count is 10,000/mm³, and serum total bilirubin concentration is 6 mg/dL with a direct component of 5 mg/dL. Ultrasonography of the right upper quadrant of the abdomen shows mild distention of the gallbladder with no gallstones. A CT scan of the abdomen shows a 5-cm cystic structure medial to the gallbladder with moderate dilation of the proximal intrahepatic ducts. Which of the following is the most appropriate next step in management?

- ☐ A) Cholescintigraphy
- ☐ B) Cholecystectomy with exploration of the common bile duct
- ☐ C) Endoscopic sphincterotomy
- ☐ D) Roux-en-Y cystojejunostomy
- ☐ E) Surgical excision of the cyst



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Following a high-speed motor vehicle collision, a 23-year-old man is brought to the emergency department. He was the unrestrained driver. His blood pressure is 150/90 mm Hg. He has retrosternal and interscapular chest pain, dyspnea, and hoarseness. X-rays show fractures of the sternum and left first rib, widening of the superior mediastinum, and caudal displacement of the left main bronchus. Which of the following is the most likely diagnosis?

- ☐ A) Flail sternum
- ☐ B) Pulmonary contusion
- ☐ C) Rupture of the esophagus
- ☐ D) Rupture of the intrathoracic trachea
- ☐ E) Rupture of the thoracic aorta



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A 62-year-old man comes to the physician 5 months after noticing a mass in the right side of his neck. He has a history of an enlarged thymus treated with radiation therapy during infancy. Examination shows a 4-cm, firm, nontender mass in the thyroid gland that moves with swallowing. Examination shows no other abnormalities. His serum thyroid-stimulating hormone concentration is within normal limits. Ultrasound-guided fine-needle aspiration biopsy shows a follicular neoplasm. Which of the following is the most appropriate next step in management?

- ☐ A) Thyroid function tests
- ☐ B) Thyroid scan
- ☐ C) Treatment with radioactive iodine
- ☐ D) Repeat fine-needle aspiration biopsy
- ☐ E) Thyroidectomy



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A previously healthy 27-year-old man comes to the physician because of a 3-month history of intermittent mild abdominal pain and diarrhea. He has had a 4.5-kg (10-lb) weight loss during this period. His temperature is 38°C (100.4°F), pulse is 72/min, and blood pressure is 114/82 mm Hg. Abdominal examination shows diffuse right lower quadrant tenderness. His hematocrit is 31%, leukocyte count is 12,700/mm³, and serum albumin concentration is 3 g/dL. The result of a barium enema is shown. Which of the following is the most likely diagnosis?

- ☐ A) Appendicitis
- ☐ B) Colon cancer
- ☐ C) Crohn disease
- ☐ D) Diverticulitis
- ☐ E) Gastroenteritis



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A 75-year-old man is scheduled to undergo elective repair of an abdominal aortic aneurysm. He has a history of hypertension controlled with drug therapy. Examination shows no abnormalities except for a palpable, 6-cm aortic aneurysm. An ECG shows normal findings. Which of the following studies is most appropriate to predict this patient's risk for perioperative myocardial infarction?

- ☐ A) Measurement of serum cholesterol concentration
- ☐ B) 24-Hour ambulatory ECG monitoring
- ☐ C) Coronary angiography
- ☐ D) Surface echocardiography
- ☐ E) Radionuclide scan with thallium and dipyridamole

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A 42-year-old man has had a 4 × 4-cm, painless ulcer over the right medial malleolus for 6 months. He has a history of deep venous thrombosis that occurred while he was hospitalized for a fracture of the right femur 20 years ago. Pedal pulses are palpable. There is 2+ edema of the right lower extremity. The left lower extremity is normal. Which of the following is the most likely cause of this ulcer?

- ☐ A) Cellulitis
- ☐ B) Lymphedema
- ☐ C) Peripheral arterial vascular disease
- ☐ D) Previous fracture and soft-tissue injury
- ☐ E) Venous valvular insufficiency

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A 27-year-old man is brought to the emergency department 40 minutes after being thrown from his motorcycle into a telephone pole. He was wearing a helmet. On arrival, he is alert and oriented to person, place, and time. His temperature is 36.8°C (98.2°F), pulse is 110/min, respirations are 20/min, and blood pressure is 100/60 mm Hg. Abdominal examination shows mild distention. The pelvis is unstable. There is blood at the urethral meatus. On digital rectal examination, the prostate is not palpable. Which of the following is the most appropriate next step in evaluation of the urethral bleeding?

- ☐ A) Abdominal ultrasonography
- ☐ B) Transrectal ultrasonography
- ☐ C) CT scan of the abdomen
- ☐ D) Cystoscopy
- ☐ E) Intravenous pyelography
- ☐ F) Retrograde urethrography



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A 37-year-old man comes to the physician because he and his wife have been unable to conceive for 3 years. They have unprotected sexual intercourse four to five times weekly. His wife has two healthy children from a previous marriage. On examination, the patient appears healthy and well developed. An irregular, ropy mass is palpated in the upper left hemiscrotum. The remainder of the examination shows no abnormalities. Results of a semen analysis are within the reference ranges. Which of the following is the most likely cause of this patient's infertility?

- ☐ A) Epididymitis
- ☐ B) Inguinal hernia
- ☐ C) Prostatitis
- ☐ D) Undescended testicle
- ☐ E) Varicocele

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Immediately after birth, a 900-g (2-lb) newborn has difficulty breathing and requires intubation and mechanical ventilation. She was born at 30 weeks' gestation to a 15-year-old mother; pregnancy and delivery were uncomplicated. Examination of the newborn shows a harsh to-and-fro murmur consistent with patent ductus arteriosus. Pharmacologic manipulation is recommended to close the patent ductus arteriosus. Indomethacin is administered, and the murmur soon disappears. Which of the following is the most likely explanation for this patient's response to the indomethacin?

- ☐ A) Cortisol inhibition with decreased norepinephrine release
- ☐ B) Cyclo-oxygenase inhibition with increased norepinephrine release
- ☐ C) Decreased arterial oxygen tension flowing through the ductus
- ☐ D) Interleukin 2 receptor blockade to promote ductal closure
- ☐ E) Secretion of surfactant by pulmonary alveolar cells

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A 35-year-old woman with chronic renal failure sustains a compression fracture of T12 after falling from a curb 2 feet from the ground. Serum studies show:

Ca ²⁺	7 mg/dL
Phosphorus	11 mg/dL
Creatinine	6 mg/dL
Alkaline phosphatase	600 U/L

Which of the following is the most likely cause of this patient's fracture?

- ☐ A) Aluminum toxicity
- ☐ B) Metastatic breast cancer
- ☐ C) Osteitis deformans (Paget disease)
- ☐ D) Primary hyperparathyroidism
- ☐ E) Secondary hyperparathyroidism

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A 42-year-old woman comes to the physician because of a 1-week history of yellow eyes, nausea, and right-sided abdominal cramps. Two months ago, she underwent a laparoscopic cholecystectomy for acute cholecystitis. Intra-operative cholangiography was not performed. She takes no medications. Vital signs are within normal limits. Examination shows scleral icterus. Cardiopulmonary examination shows no abnormalities. Abdominal examination shows right upper quadrant tenderness with no distention or peritoneal signs. Bowel sounds are normal. Serum studies show:

Bilirubin, total	8 mg/dL
Direct	5 mg/dL
Alkaline phosphatase	650 U/L
AST	20 U/L
ALT	18 U/L

Which of the following is the most likely cause of this patient's jaundice?

- ☐ A) Choledocholithiasis
- ☐ B) Cirrhosis
- ☐ C) Pancreatic cancer
- ☐ D) Postoperative bile leak
- ☐ E) Primary biliary sclerosis

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A previously healthy 19-year-old woman is brought to the emergency department because of a 4-hour history of constant severe abdominal pain. Her symptoms began 3 weeks ago as intermittent, colicky pain; examination 1 week ago showed a 5-cm right ovarian cyst. Abdominal examination now shows tenderness with rebound and guarding in the right lower quadrant. There is a palpable, tender, 10-cm right adnexal mass. X-rays of the chest and abdomen show no abnormalities. Pelvic ultrasonography confirms a complex, cystic mass. Her leukocyte count is $12,000/\text{mm}^3$. Urinalysis shows no abnormalities. A pregnancy test is negative. Which of the following is the most appropriate next step in management?

- ☐ A) Admission to the hospital for observation
- ☐ B) Antibiotic therapy
- ☐ C) Colposcopy
- ☐ D) CT scan-guided drainage
- ☐ E) Exploratory operation

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Nine days after undergoing pylorus-sparing pancreaticoduodenectomy for adenocarcinoma of the pancreas, a 62-year-old man has dyspnea. A nasogastric tube is draining 500 mL of fluid daily. Over the past 3 days, an operative drain in the right upper quadrant of the abdomen has been draining 200 mL to 300 mL of clear fluid daily. Analysis of the fluid shows an amylase activity greater than 5000 U/L. His pulse is 90/min, respirations are 24/min, and blood pressure is 130/70 mm Hg. Serum studies show:

Na ⁺	138 mEq/L
Cl ⁻	112 mEq/L
K ⁺	4.2 mEq/L
HCO ₃ ⁻	18 mEq/L

Arterial blood gas analysis on room air:

pH	7.32
Pco ₂	22 mm Hg
Po ₂	95 mm Hg

Which of the following is the most likely mechanism of these findings?

- ☐ A) Adrenal insufficiency
- ☐ B) Inadequate renal blood flow
- ☐ C) Increased lactic acid production
- ☐ D) Injury to the renal tubules
- ☐ E) Loss of bicarbonate from the gastrointestinal tract
- ☐ F) Loss of bicarbonate through the kidneys
- ☐ G) Nasogastric suction

Over the past 6 weeks, a 30-year-old woman has had increasing nervousness, sweating, and difficulty sleeping; she has had a 4.5-kg (10-lb) weight loss during this period. She has noted a lump on the right side of the neck. Her pulse is 110/min, and blood pressure is 130/50 mm Hg. Examination shows a 4 × 4-cm mass on the right side of the thyroid gland; no other thyroid tissue is palpated. There are no bruits or regional adenopathy. Thyroid scan shows uptake only in the area of the nodule. Which of the following is the most likely diagnosis?

- ☐ A) Chronic lymphocytic thyroiditis (Hashimoto disease)
- ☐ B) Follicular carcinoma of the thyroid gland
- ☐ C) Graves disease
- ☐ D) Papillary thyroid carcinoma
- ☐ E) Toxic thyroid adenoma

Five days after undergoing incision and drainage of a left inguinal sebaceous cyst, a 57-year-old man comes to the emergency department because of an 8-hour history of moderate scrotal pain. He has type 2 diabetes mellitus, coronary artery disease, and hypercholesterolemia. His medications are a quinolone, sulfonylurea, β -adrenergic blocking agent, and statin. His temperature is 38.9°C (102°F), pulse is 120/min and regular, and blood pressure is 90/50 mm Hg. Examination of the scrotum shows edema and tenderness at the base; there is a 3-cm black surgical incision with surrounding erythema extending to and including the site of the sebaceous cyst. Laboratory studies show:

Leukocyte count	21,000/mm ³
Segmented neutrophils	70%
Bands	7%
Eosinophils	3%
Lymphocytes	20%
Serum	
Na ⁺	130 mEq/L
K ⁺	3.4 mEq/L
Cl ⁻	94 mEq/L
HCO ₃ ⁻	17 mEq/L
Glucose	324 mg/dL

Intravenous administration of 0.9% saline and insulin is begun. Which of the following is the most appropriate next step in management?

- ☐ A) Application of warm compresses to the operative site
- ☐ B) Hyperbaric oxygen therapy
- ☐ C) Initiation of intravenous antifungal therapy
- ☐ D) Initiation of intravenous potassium therapy
- ☐ E) Surgical debridement

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Surgery Self-Assessment

A 27-year-old woman is brought to the emergency department 30 minutes after she sustained a deep, 10-cm laceration to her back. Infiltration of which of the following anesthetic agents will provide the longest duration of analgesia for wound management in this patient?

- ☐ A) Bupivacaine
- ☐ B) Chloroprocaine
- ☐ C) Lidocaine
- ☐ D) Prilocaine
- ☐ E) Procaine

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During exploratory celiotomy, a 4 × 7-cm cystic lesion in the tail of the pancreas is found in a 46-year-old woman. Biopsy of the cyst wall shows cystadenoma. Which of the following is the most appropriate management of this lesion?

- ☐ A) Internal drainage into a Roux-en-Y limb of the jejunum
- ☐ B) Simple external drainage
- ☐ C) Distal pancreatectomy
- ☐ D) Total pancreatectomy
- ☐ E) Whipple operation

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A 67-year-old man is admitted to the hospital because of a 6-week history of intermittent nausea and vomiting. He has had decreased appetite resulting in a 16-kg (35-lb) weight loss during this period. He underwent a distal gastrectomy for peptic ulcer disease 35 years ago. He appears cachectic. He is 173 cm (5 ft 8 in) tall and weighs 50 kg (110 lb); BMI is 17 kg/m². His temperature is 37.2°C (99°F), pulse is 100/min, and blood pressure is 110/80 mm Hg. Examination shows severe temporal wasting. The abdomen is scaphoid; there is mild epigastric tenderness. The stool is brown, and test for occult blood is negative. Upper gastrointestinal endoscopy shows a mass filling the remaining portion of the stomach. A biopsy specimen of the mass shows adenocarcinoma. This patient is most likely to have which of the following metabolic abnormalities?

- ☐ A) Decreased lipolysis
- ☐ B) Decreased serum interleukin-6 concentration
- ☐ C) Hypertriglyceridemia
- ☐ D) Hypoglycemia
- ☐ E) Increased serum tumor necrosis factor concentration

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A 24-year-old woman is scheduled to undergo laparoscopy for evaluation of chronic pelvic pain. Four days prior to the procedure, informed consent is obtained by the physician and witnessed by her husband. On the morning of the operation, she receives a narcotic premedication. Five minutes later, she becomes panicky, stating that she is afraid and does not wish to undergo the scheduled operation. Which of the following is the most appropriate intervention?

- ☐ A) Cancel the operation
- ☐ B) Obtain the opinion of a psychiatrist regarding her competence to make decisions regarding her medical care
- ☐ C) Proceed with the operation after administering a narcotic antagonist
- ☐ D) Proceed with the operation as planned
- ☐ E) Reconfirm with her husband that she was not under the influence of narcotics when the patient signed the consent, treat the panic attack with a benzodiazepine, and proceed with the operation



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Two days after undergoing a left carotid endarterectomy, a 62-year-old man is brought to the emergency department because of slurred speech and weakness of the right arm and leg for 1 hour. On discharge yesterday, he had a mild, persistent left-sided headache, but neurologic examination showed no focal findings. He currently takes one aspirin daily. He appears anxious and diaphoretic. His pulse is 95/min, respirations are 18/min, and blood pressure is 170/95 mm Hg. Examination shows a clean, dry, and intact left neck incision without evidence of swelling. Cranial nerves are intact. Muscle strength is 2/5 on the right and 5/5 on the left. His speech is slurred, and he is unable to name simple objects. Carotid duplex ultrasonography shows no abnormalities. A CT scan of the head shows an intraparenchymal hemorrhage. This complication is most likely a result of which of the following?

- ☐ A) Carotid thrombosis
- ☐ B) Hypertension
- ☐ C) Intracranial aneurysm
- ☐ D) Intracranial tumor
- ☐ E) Platelet dysfunction

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Surgery Self-Assessment

A 52-year-old woman comes to the physician because of three episodes of bright red bleeding during bowel movements during the past 2 weeks. She has a 10-year history of hypercholesterolemia treated with simvastatin. Her temperature is 36.8°C (98.2°F), pulse is 88/min, respirations are 18/min, and blood pressure is 139/82 mm Hg. Cardiopulmonary examination shows no abnormalities. The abdomen is soft and nontender; no masses are palpated. Rectal examination shows internal hemorrhoids that prolapse and are manually reducible. Her hemoglobin concentration is 13 g/dL, leukocyte count is 6500/mm³, and serum cholesterol concentration is 210 mg/dL. Which of the following is the most appropriate next step in management?

- ☐ A) CT scan of the abdomen
- ☐ B) Technetium 99m scan
- ☐ C) Docusate sodium therapy
- ☐ D) Colonoscopy
- ☐ E) Hemorrhoidectomy



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An 18-year-old man is brought to the emergency department 10 minutes after sustaining a stab wound to the right chest. On arrival, he has shortness of breath. His pulse is 120/min, and blood pressure is 70/40 mm Hg. Breath sounds are absent on the right. The trachea is midline. The neck veins are flat. Which of the following is the most likely diagnosis?

- ☐ A) Lobar atelectasis
- ☐ B) Massive hemothorax
- ☐ C) Pulmonary embolism
- ☐ D) Ruptured bronchus
- ☐ E) Tension pneumothorax



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A previously healthy 42-year-old man comes to the physician because of a 2-day history of right knee pain and an inability to extend the right knee. The symptoms began when he was getting up from a low chair. His temperature is 37°C (98.6°F). Examination of the knee shows tenderness to palpation along the medial joint line and a joint effusion. Ligament stability is normal. Range of motion is from 15 to 110 degrees. An x-ray of the knee shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Anterior cruciate ligament injury
- ☐ B) Bursitis
- ☐ C) Chondromalacia
- ☐ D) Collateral ligament injury
- ☐ E) Patellar dislocation
- ☐ F) Patellar tendon rupture
- ☐ G) Posterior cruciate ligament injury
- ☐ H) Torn meniscus



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A 67-year-old man comes to the physician because of a 1-week history of increasingly severe abdominal pain, fever, vomiting, and decreased appetite. Two years ago, he was admitted to the hospital for a similar episode. He has had intermittent left lower abdominal pain during the past 5 years. His temperature is 39.6°C (103.3°F), pulse is 105/min, respirations are 22/min, and blood pressure is 150/90 mm Hg. Abdominal examination shows left lower quadrant tenderness with rebound. The remainder of the abdomen is soft and nontender. No masses are palpated on rectal examination. Test of the stool for occult blood is negative. His leukocyte count is 21,000/mm³ with a left shift. X-rays of the abdomen show a nonspecific gas pattern. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Intravenous pyelography
- ☐ B) Visceral angiography
- ☐ C) CT scan of the abdomen and pelvis
- ☐ D) Colonoscopy
- ☐ E) Exploratory laparotomy



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The response options for the next 2 items are the same. Select one answer for each item in the set.

For each patient with an ophthalmologic abnormality, select the most likely diagnosis.

- | | |
|---|---|
| ☐ A) Central retinal artery occlusion | ☐ J) Iridocyclitis |
| ☐ B) Central retinal vein occlusion | ☐ K) Lens subluxation |
| ☐ C) Chalazion | ☐ L) Optic neuritis |
| ☐ D) Corneal abrasion | ☐ M) Retinal detachment |
| ☐ E) Functional blindness | ☐ N) Retrobulbar hemorrhage |
| ☐ F) Glaucoma | ☐ O) Scleral rupture |
| ☐ G) Herpes simplex conjunctivitis | ☐ P) Temporal arteritis |
| ☐ H) Hordeolum | ☐ Q) Traumatic mydriasis and miosis |
| ☐ I) Hyphema | ☐ R) Vitreous hemorrhage |

An 8-year-old girl has had the sensation that there is something in her left eye for 1 day and photophobia and decreased vision for 6 hours. Her parents have noticed that she has been frequently rubbing her left eye. Examination shows increased tearing with erythema of the conjunctiva. There are small vesicles with an erythematous base on the eyelid skin and the conjunctiva.

For each patient with an ophthalmologic abnormality, select the most likely diagnosis.

- | | |
|---|---|
| ☐ A) Central retinal artery occlusion | ☐ J) Iridocyclitis |
| ☐ B) Central retinal vein occlusion | ☐ K) Lens subluxation |
| ☐ C) Chalazion | ☐ L) Optic neuritis |
| ☐ D) Corneal abrasion | ☐ M) Retinal detachment |
| ☐ E) Functional blindness | ☐ N) Retrobulbar hemorrhage |
| ☐ F) Glaucoma | ☐ O) Scleral rupture |
| ☐ G) Herpes simplex conjunctivitis | ☐ P) Temporal arteritis |
| ☐ H) Hordeolum | ☐ Q) Traumatic mydriasis and miosis |
| ☐ I) Hyphema | ☐ R) Vitreous hemorrhage |

A 66-year-old woman had the acute onset of loss of vision in her right eye 1 hour ago. There is no pain or any other associated symptoms, and she has no history of previous eye problems or allergies. She has type 2 diabetes mellitus and hypertension treated with hydrochlorothiazide. Examination shows blindness in the right eye. The pupil does not react to direct light but consensually reacts to light in the opposite pupil. Ocular movement is intact in the affected eye. Ophthalmoscopic examination shows pallor of the optic disc, macular edema, thin arterioles, and sausage-like narrowing of the retinal veins.

A 68-year-old man has had recurrent stridor for 2 hours. Two years ago, he underwent radiation therapy for laryngeal cancer. Examination shows a bulky tumor involving the upper and middle neck bilaterally. Arterial blood gas analysis while breathing 100% oxygen shows:

pH	7.32
Pco ₂	52 mm Hg
PO ₂	55 mm Hg
HCO ₃ ⁻	17 mEq/L

Which of the following is the most appropriate next step in management?

- ☐ A) Placement of an esophageal airway
- ☐ B) Bronchoscopy
- ☐ C) Neck irradiation
- ☐ D) Tracheostomy
- ☐ E) Laryngectomy

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A 37-year-old primigravid woman at 26 weeks' gestation comes to the physician for a routine prenatal visit. She has a 12-month history of pain of the left wrist with occasional numbness of the palm. She uses a computer keyboard every day at work and at home. Her symptoms were previously alleviated with regular breaks from keyboard use and occasional use of ibuprofen but have become more difficult to control during her pregnancy. Her pregnancy has been otherwise uncomplicated. There is tingling of the index and middle fingers of the left hand with percussion of the volar wrist. The remainder of the examination shows no abnormalities. Pelvic examination shows a uterus consistent in size with a 26-week gestation. Which of the following is most likely to confirm the diagnosis?

- ☐ A) X-rays of the hands and wrists
- ☐ B) Electromyography
- ☐ C) MRI of the hands and wrists
- ☐ D) Arthroscopy of the wrist joints
- ☐ E) Nerve conduction studies



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An 87-year-old man comes to the physician as a new patient. He has had three episodes of pneumonia over the past 6 months; his last episode 2 months ago required treatment in the hospital. His wife reports that over the past year he has had increasingly bad breath. He appears well but has halitosis. The remainder of the examination shows no abnormalities. A complete blood count, serum electrolyte concentrations, and liver function tests show no abnormalities. An x-ray of the chest shows scarring at the right lung base. Which of the following is the most likely diagnosis?

- ☐ A) Achalasia
- ☐ B) Carcinoma of the cervical esophagus
- ☐ C) Carcinoma of the right lung
- ☐ D) Cerebral infarction with bulbar palsy
- ☐ E) Pharyngoesophageal (Zenker) diverticulum



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National Board of Medical Examiners
Surgery Self-Assessment

A 37-year-old man is brought to the emergency department 30 minutes after being involved in a motor vehicle collision. On arrival, he reports abdominal pain. His pulse is 84/min, respirations are 18/min, and blood pressure is 110/82 mm Hg. Abdominal examination shows diffuse tenderness. Bowel sounds are decreased. A supine x-ray of the chest is shown. Which of the following is the most likely diagnosis?

- ☐ A) Gastric perforation
- ☐ B) Lung abscess
- ☐ C) Pneumopericardium
- ☐ D) Pneumothorax
- ☐ E) Ruptured diaphragm



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National Board of Medical Examiners
Surgery Self-Assessment

A 56-year-old woman has a 3-month history of excessive sweating, nervousness, tremors, tachycardia, and hunger. Her symptoms are worsened by fasting. Selective angiography shows a 1-cm tumor in the tail of the pancreas. Which of the following is the most likely diagnosis?

- ☐ A) Carcinoid
- ☐ B) Glucagonoma
- ☐ C) Insulinoma
- ☐ D) Somatostatinoma
- ☐ E) Vipoma

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National Board of Medical Examiners
Surgery Self-Assessment

A 77-year-old resident of a skilled nursing care facility is brought to the emergency department because of fever and vomiting for the past 2 days. She is alert but is unable to give a history. She asks repeatedly for a drink of water. Her temperature is 38.6°C (101.5°F), and blood pressure is 100/60 mm Hg. Examination shows a distended, nontender abdomen with sparse high-pitched bowel sounds. A supine x-ray of the abdomen shows multiple dilated loops of small bowel and gas within the small-bowel lumen and within the liver. Which of the following is the most likely cause of these findings?

- ☐ A) Bacterial cholangitis caused by *Klebsiella pneumoniae*
- ☐ B) Cholecystoduodenal fistula with an impacted gallstone
- ☐ C) Emphysematous cholecystitis with an intrahepatic perforation
- ☐ D) Perforated duodenal ulcer with a subhepatic abscess
- ☐ E) Pylephlebitis caused by sigmoid diverticulitis



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National Board of Medical Examiners
Surgery Self-Assessment

A 57-year-old man comes to the physician because of a 2-month history of difficulty swallowing. He says that food gets stuck when he swallows, and he has been drinking more water to help with this. He has moderate discomfort when he has to make an effort to swallow. He has no history of serious illness and takes no medications. He has smoked two packs of cigarettes daily for 40 years. He has a long-standing history of drinking five alcoholic beverages daily. Examination shows no abnormalities. A chest x-ray shows no abnormalities. Which of the following is the most appropriate next step in management?

- ☐ A) CT scan of the chest
- ☐ B) CT scan of the neck and chest
- ☐ C) Two-month trial of high doses of omeprazole
- ☐ D) Esophageal manometry
- ☐ E) Esophagogastroduodenoscopy

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An 80-year-old woman with atherosclerosis and chronic atrial fibrillation has had severe midabdominal pain for 4 hours. Examination shows mild abdominal tenderness and absent bowel sounds; there are no masses or signs of peritoneal irritation. Test of the stool for occult blood is positive. Leukocyte count is $28,340/\text{mm}^3$ with 65% segmented neutrophils, 20% bands, and 15% lymphocytes. Arterial blood gas analysis on room air shows:

pH	7.18
Pco ₂	35 mm Hg
Po ₂	62 mm Hg

An x-ray of the abdomen shows a nonspecific gas pattern. Which of the following is the most appropriate next step in management?

- ☐ A) Colonoscopy
- ☐ B) Exploratory celiotomy
- ☐ C) Gastroscopy
- ☐ D) Laparoscopy
- ☐ E) Peritoneal lavage

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National Board of Medical Examiners
Surgery Self-Assessment

A 67-year-old woman has the onset of dyspnea, myalgias, and flank pain during transfusion of a second unit of packed red blood cells immediately after a colon resection for colon cancer. Intraoperative blood loss was 1200 mL. There is blood in the surgical drain, and the urine is dark. Her postoperative hematocrit was 21%. Which of the following tests is most likely to establish the diagnosis?

- ☐ A) Aerobic and anaerobic blood cultures
- ☐ B) Arterial blood gas analysis
- ☐ C) Measurement of hemoglobin concentration
- ☐ D) Repeat typing and crossmatching of the transfused blood
- ☐ E) Screening for disseminated intravascular coagulation

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A 42-year-old woman comes to the physician for a routine examination. She has a 10-year history of intermittent palpitations. Her symptoms last a few minutes and are not accompanied by chest pain, shortness of breath, or light-headedness; they are exacerbated by stress and caffeine. She has never taken her pulse during these episodes. She is an avid rock climber and mountain biker and does not recall any episodes during these activities. She does not smoke or drink alcohol. She is 173 cm (5 ft 8 in) tall and weighs 61 kg (135 lb); BMI is 21 kg/m². Her pulse is 64/min and regular, and blood pressure is 118/76 mm Hg. The lungs are clear to auscultation. Cardiac examination shows a midsystolic click heard best at the apex. The abdomen is scaphoid with normal bowel sounds. A resting ECG shows no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Atrial fibrillation
- ☐ B) Hyperthyroidism
- ☐ C) Mitral regurgitation
- ☐ D) Mitral valve prolapse
- ☐ E) Panic attack
- ☐ F) Premature ventricular contractions



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A 57-year-old woman comes to the emergency department because of a 1-day history of fever, chills, and generalized weakness. She underwent splenectomy for immune thrombocytopenic purpura 10 years ago. At that time, she received 4 units of packed red blood cells and all recommended vaccines. She has not seen a physician for 9 years. She takes no medications. She does not smoke or drink alcohol. She appears weak but answers questions appropriately. Her temperature is 39.3°C (102.8°F), pulse is 100/min, respirations are 20/min, and blood pressure is 80/50 mm Hg. Crackles are heard at the right lung base. A grade 1/6 systolic murmur is heard best at the upper left sternal border. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hemoglobin	12.5 g/dL
Leukocyte count	18,500/mm ³
Segmented neutrophils	85%
Bands	10%
Lymphocytes	5%
Platelet count	125,000/mm ³

A blood smear shows Howell-Jolly bodies. An x-ray of the chest shows a consolidation in the right lower lobe of the lung. In addition to intravenous administration of fluids, which of the following is the most appropriate next step in management?

- ☐ A) Ceftriaxone and vancomycin therapy
- ☐ B) Imipenem therapy
- ☐ C) Quinine and clindamycin therapy
- ☐ D) Trimethoprim-sulfamethoxazole therapy
- ☐ E) Plasmapheresis

A 67-year-old postal worker comes to the physician 1 day after a 5-minute episode of weakness and numbness in his right (dominant) hand while at work. He did not have visual problems, headache, or weakness or numbness of the lower extremities. He is currently asymptomatic. He has smoked one pack of cigarettes daily for 45 years. He has not seen a physician in 40 years. His pulse is 85/min and irregular, and blood pressure is 140/90 mm Hg. The lungs are clear to auscultation. Pulses are palpable. Carotid bruits are heard bilaterally. Heart sounds are normal except for frequent premature beats. Neurologic examination shows no abnormalities. An ECG shows a normal sinus rhythm with multiple atrial premature contractions. Which of the following is the most appropriate next step in management?

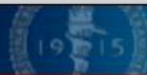
- ☐ A) Echocardiography
- ☐ B) Carotid duplex ultrasonography
- ☐ C) Heparin therapy
- ☐ D) Tissue plasminogen activator therapy
- ☐ E) Warfarin therapy



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Surgery Self-Assessment

A 76-year-old man sustained a fracture of the midshaft of the right femur during a motor vehicle collision. In the emergency department, the circumference of his right thigh is two times greater than the left. His pulse is 120/min and regular, respirations are 16/min, and blood pressure is 80/60 mm Hg. A large-bore intravenous catheter is inserted for fluid and blood infusion, and a Foley catheter is inserted. Which of the following findings best indicates adequate resuscitation in this patient?

- ☐ A) Decreased pulse to 90/min
- ☐ B) Decreased respirations to 12/min
- ☐ C) Increased systolic blood pressure to 100 mm Hg or greater
- ☐ D) Urine output of 30–40 mL/h
- ☐ E) Normal capillary filling pressure



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Surgery Self-Assessment

A 28-year-old woman comes to the physician for a routine examination. She has been in excellent health and takes no medications. She has recently gained 36 kg (80 lb), which she attributes to stress associated with a divorce. She is 163 cm (5 ft 4 in) tall and weighs 106 kg (235 lb); BMI is 40 kg/m². Blood pressure is 130/85 mm Hg. Her obesity places her at greatest risk for developing which of the following conditions over the next 5 years?

- ☐ A) Cholecystitis
- ☐ B) Diverticulosis
- ☐ C) Irritable bowel syndrome
- ☐ D) Pancreatitis
- ☐ E) Peptic ulcer disease



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Surgery Self-Assessment

A 27-year-old man is brought to the emergency department 25 minutes after a motor vehicle collision in which he was the unrestrained driver. On arrival, his Glasgow Coma Scale score is 5. His pulse is 150/min, respirations are 15/min and shallow, and blood pressure is 165/90 mm Hg. The right pupil is 6 mm, and the left pupil is 4 mm. There is minimal response to noxious stimuli. Which of the following is the most appropriate next step in management?

- ☐ A) CT scan of the head
- ☐ B) Intravenous administration of labetalol
- ☐ C) Intubation and hyperventilation
- ☐ D) EEG
- ☐ E) Lumbar puncture

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Exam Section

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Surgery Self-Assessment

An otherwise healthy 30-year-old man has had an acneiform infection of the left axilla for 2 weeks. He has had previous infections in the same region and groin area. Examination shows acute inflammatory changes with purulent drainage from small sinus openings along the indurated skin of the axilla. There are multiple sinus openings along both groins and the right axilla. This infection most likely involves which of the following?

- ☐ A) Apocrine glands
- ☐ B) Eccrine glands
- ☐ C) Hair follicles
- ☐ D) Lymph nodes
- ☐ E) Sebaceous glands

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National Board of Medical Examiners
Surgery Self-Assessment

While receiving radiation therapy for a glioma in the right parietotemporal area, a 62-year-old man has the onset of severe abdominal pain. He is being treated with corticosteroids for cerebral edema and intravenous fluids for nausea and vomiting. Bowel sounds are minimal. There is tenderness in all four quadrants of the abdomen. An x-ray of the abdomen shows free air under the diaphragm. Which of the following is the most appropriate next step in management?

- ☐ A) Upper gastrointestinal series with soluble contrast
- ☐ B) Water-soluble contrast enema
- ☐ C) CT scan of the head
- ☐ D) Upper endoscopy
- ☐ E) Laparotomy



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A 47-year-old man has had the pearly surfaced papule shown on the upper lip for 18 months. He is otherwise asymptomatic. Examination shows some telangiectatic matting over the lesion. Which of the following is the most likely diagnosis?

- ☐ A) Basal cell carcinoma
- ☐ B) Blue nevus
- ☐ C) Epidermoid cyst
- ☐ D) Keratoacanthoma
- ☐ E) Melanoma
- ☐ F) Seborrheic keratosis
- ☐ G) Squamous cell carcinoma



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Surgery Self-Assessment

A previously healthy 15-year-old boy has the sudden onset of anterior right-sided chest pain. His temperature is 36.9°C (98.4°F), pulse is 90/min, respirations are 18/min, and blood pressure is 110/70 mm Hg. Examination shows no abnormalities. X-rays of the chest show a small, right pneumothorax. Which of the following is the most appropriate next step in evaluation?

- ☐ A) Observation
- ☐ B) Pulmonary function tests
- ☐ C) CT scan of the chest
- ☐ D) Ventilation-perfusion lung scans
- ☐ E) Bronchoscopy



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Surgery Self-Assessment

A 72-year-old woman is brought to the emergency department 48 hours after slipping on a rug and falling. Her son found her lying on the floor. On arrival, she has severe pain of the right hip and thigh. Her pulse is 126/min, respirations are 30/min, and blood pressure is 80/40 mm Hg. There is marked external rotation and shortening of the right lower extremity. Pedal pulses are present. There is a normal sinus rhythm. Which of the following is the most likely cause of her hemodynamic status?

- ☐ A) Fat embolism
- ☐ B) Hypovolemia
- ☐ C) Myocardial infarction
- ☐ D) Pulmonary embolism
- ☐ E) Reaction to pain



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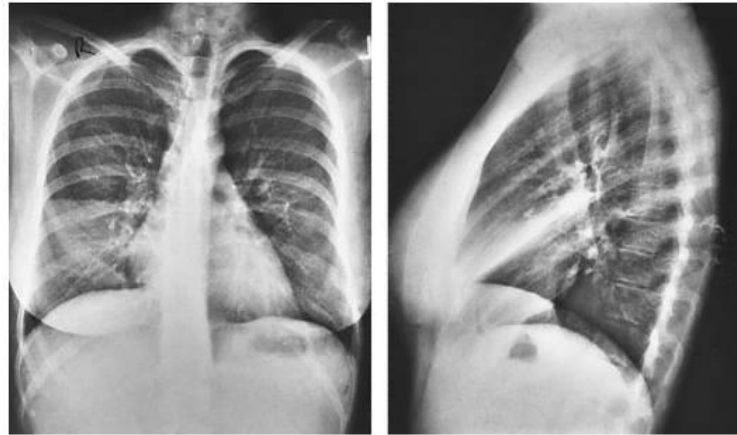
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A hospitalized 15-year-old girl has had progressive shortness of breath for 2 hours. She underwent an appendectomy yesterday and has not been out of bed since then. She has had a mild dry cough but no chest pain. Two liters of 0.9% saline have been administered over the past 24 hours. She is alert. Her temperature is 37.7°C (99.8°F), pulse is 90/min and regular, respirations are 22/min, and blood pressure is 105/64 mm Hg. There is no jugular venous distention. Breath sounds are decreased. There is dullness to percussion over the right midlung field with egophony. No wheezes are heard. A grade 3/6 holosystolic murmur is heard best at the left sternal border with radiation to the axilla. There are no gallops. X-rays of the chest are shown. Which of the following is the most likely underlying mechanism of this patient's shortness of breath?

- ☐ A) Acute thrombosis of the right pulmonary artery
- ☐ B) Aspiration of gastric contents during preoperative intubation
- ☐ C) Collapse of the right middle lobe of the lung from decreased inspiratory effort
- ☐ D) Postoperative infection with hospital-acquired organisms
- ☐ E) Severe mitral regurgitation with perioperative volume overload

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Surgery Self-Assessment Time Remaining:
4 hr 25 min 18 sec

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A 19-year-old man comes to the emergency department because of severe generalized abdominal pain for 12 hours. He has had increasing episodes of diarrhea for 6 weeks and bloody diarrhea for 4 days. He has had no bowel movements since the onset of abdominal pain. He appears pale and ill. His temperature is 38.9°C (102°F), and pulse is 144/min. Abdominal examination shows diffuse tenderness with hypoactive bowel sounds. Rectal examination shows bloody stool. An x-ray of the abdomen shows a transverse colon 8 cm in diameter but no free air. Which of the following is the most likely diagnosis?

- ☐ A) Angiodysplasia of the colon
- ☐ B) Antibiotic-associated colitis
- ☐ C) Arteriovenous malformation of the colon
- ☐ D) Diverticulitis
- ☐ E) Ischemic colitis
- ☐ F) Pseudomembranous colitis
- ☐ G) Rectal carcinoid
- ☐ H) Ulcerative colitis
- ☐ I) Volvulus of the colon

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